



DPUTRFDDataFull

DPUTRFDDataFull makes available 24 classifications of surrounding vehicles.

1. Outputs:

<i>SpLong</i> [TOKEN]	Longitudinal distance center of gravity / center of gravity [m]
<i>SpLat</i> [TOKEN]	Lateral distance center of gravity / center of gravity [m]
<i>StLong</i> [TOKEN]	Longitudinal distance fender / fender [m]
<i>StLat</i> [TOKEN]	Lateral distance fender / fender [m]
<i>dLong</i> [TOKEN]	Longitudinal distance center of gravity / center of gravity [m]
<i>dLat</i> [TOKEN]	Lateral distance center of gravity / center of gravity [m]
<i>v</i> [TOKEN]	Speed [m/s]
<i>X</i> [TOKEN]	X coordinate [m]
<i>Y</i> [TOKEN]	Y coordinate [m]
<i>Z</i> [TOKEN]	Z coordinate [m]
<i>yaw</i> [TOKEN]	Yaw angle [rad]
<i>pitch</i> [TOKEN]	Pitch angle [rad]
<i>roll</i> [TOKEN]	Roll angle [rad]
<i>ID</i> [TOKEN]	Vehicle ID
<i>UserID</i> [TOKEN]	Vehicle UserID
<i>Type</i> [TOKEN]	ODB type of the vehicle
<i>Length</i> [TOKEN]	Length of the vehicle [m]
<i>LengthFront</i> [TOKEN]	Length of the vehicle from center of gravity to front edge [m]
<i>LengthRear</i> [TOKEN]	Length of the vehicle from center of gravity to rear edge [m]
<i>Width</i> [TOKEN]	Width of the vehicle [m]
<i>WidthLeft</i> [TOKEN]	Width of the vehicle from center of gravity to left edge [m]
<i>WidthRight</i> [TOKEN]	Width of the vehicle from center of gravity to right edge [m]
<i>LaneID</i> [TOKEN]	Lane number of the vehicle
<i>ax</i> [TOKEN]	Acceleration set for the behavior model [m/s ²]
<i>dax</i> [TOKEN]	Calculated acceleration [m/s ²]

2. Dependencies on other DPUs:

DPUTRFDDataFull must run on a computer together with DPUTRF(X). The DPU *Index* entry for DPUTRFDDataFull must be higher than the *Index* entry for DPUTRF(X).

3. Operation:

Vehicle classification

After determination of the distance for all vehicles, the vehicles recorded are classified with reference to the EGO-vehicle.



This classification is only valid in Courses, which limits the use of DPUTRFDDataFull to Courses. In Areas the DPUTRFDDataSingle and DPUTRFTarget can be used to record data of simulated vehicles.

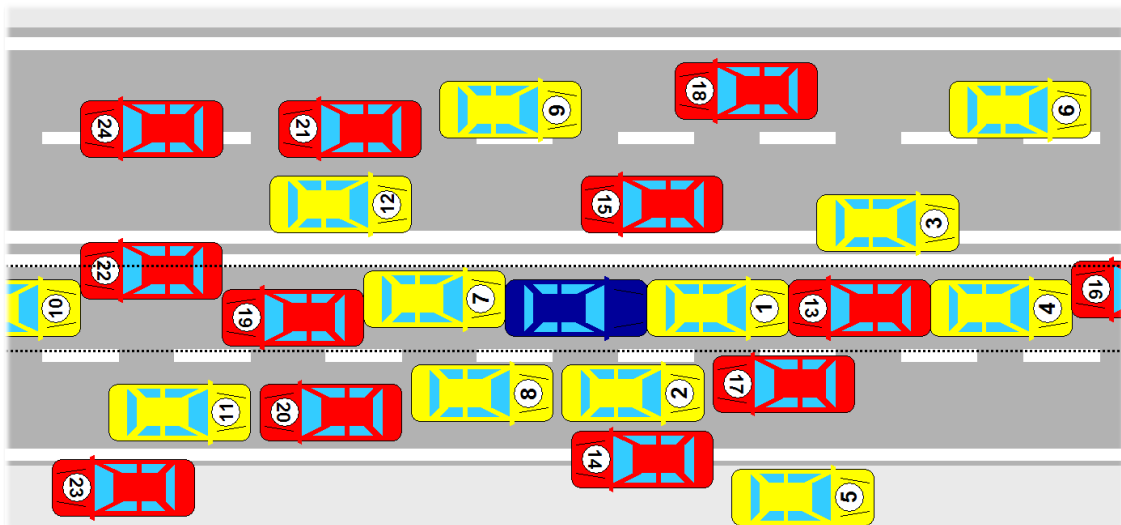


Figure 1 Vehicle classification

A check is performed as to whether the center of gravity a vehicle is located in the same lane as the center of gravity of the EGO-vehicle.

This vehicle is then classified as ahead in the same lane (1, 4, 13 and 16 in Figure 9). Vehicles outside the same lane (represented by dotted lines in Figure 9) are classified as either left or right of the lane.

This is then followed by a classification into "same direction of travel" (yellow in Figure 9) and "on-coming direction of travel" (red in Figure 9). The vehicles are now sorted according to pLong within their class, and classified as next and next but one vehicle.

Vehicle classes taking into account the direction of travel

The number specified in the second column refers to Figure 9

[Token]	No.	Description
SameAheadNext	1	
SameAheadNextRight	2	Vehicle ahead in the (same/right/left)
SameAheadNextLeft	3	lane with the same direction of travel as the EGO-vehicle
SameAheadScnd	4	Vehicle traveling in the same direction
SameAheadScndRight	5	ahead of SameAheadNextxxx
SameAheadScndLeft	6	



SILAB DRIVING SIMULATION

version Version 7.1 documentation

SameBehindNext	7	Vehicle following the EGO-vehicle in the same direction of travel as the EGO-vehicle in the (same/right/left) lane
SameBehindNextRight	8	
SameBehindNextLeft	9	
SameBehindScnd	10	Oncoming vehicle directly following the TowardsAheadXXX Vehicle
SameBehindScndRight	11	
SameBehindScndLeft	12	
TowardsAheadNext	13	Oncoming vehicle approaching in front of the EGO-vehicle in the (same/right/left) lane
TowardsAheadNextRight	14	
TowardsAheadNextLeft	15	
TowardsAheadScnd	16	Oncoming vehicle directly following the TowardsAheadXXX vehicle
TowardsAheadScndRight	17	
TowardsAheadScndLeft	18	
TowardsBehindNext	19	Vehicle behind the EGO-vehicle moving away in the (same/right/left) lane
TowardsBehindNextRight	20	
TowardsBehindNextLeft	21	
TowardsBehindScnd	22	Vehicle in front of the TowardsBehindNext vehicle
TowardsBehindScndRight	23	
TowardsBehindScndLeft	24	